TinkerSub KB-1 side plates: cutting templates for one 24 x 24 in sheet of 1/2

1 Print every page at 100 % / Actual size, NOT 'Scale to fit'. Measure the 100 mm bar on a page: it must be exactly 100 mm (about 3 15/16 in).

2 Each plate is several tiles (rows A, B, C from the top, columns 1, 2, 3 from the left). Trim a tile on its grey dashed line, overlap the next tile, line up the red circles and tape the joints on the front.

3 Lay both templates on the sheet as shown on the right, 5 mm apart (the sheet has only ~2 mm to spare at each edge: measure it first). Spray adhesive (3M Super 77) or glue stick holds them; peel off afterwards.

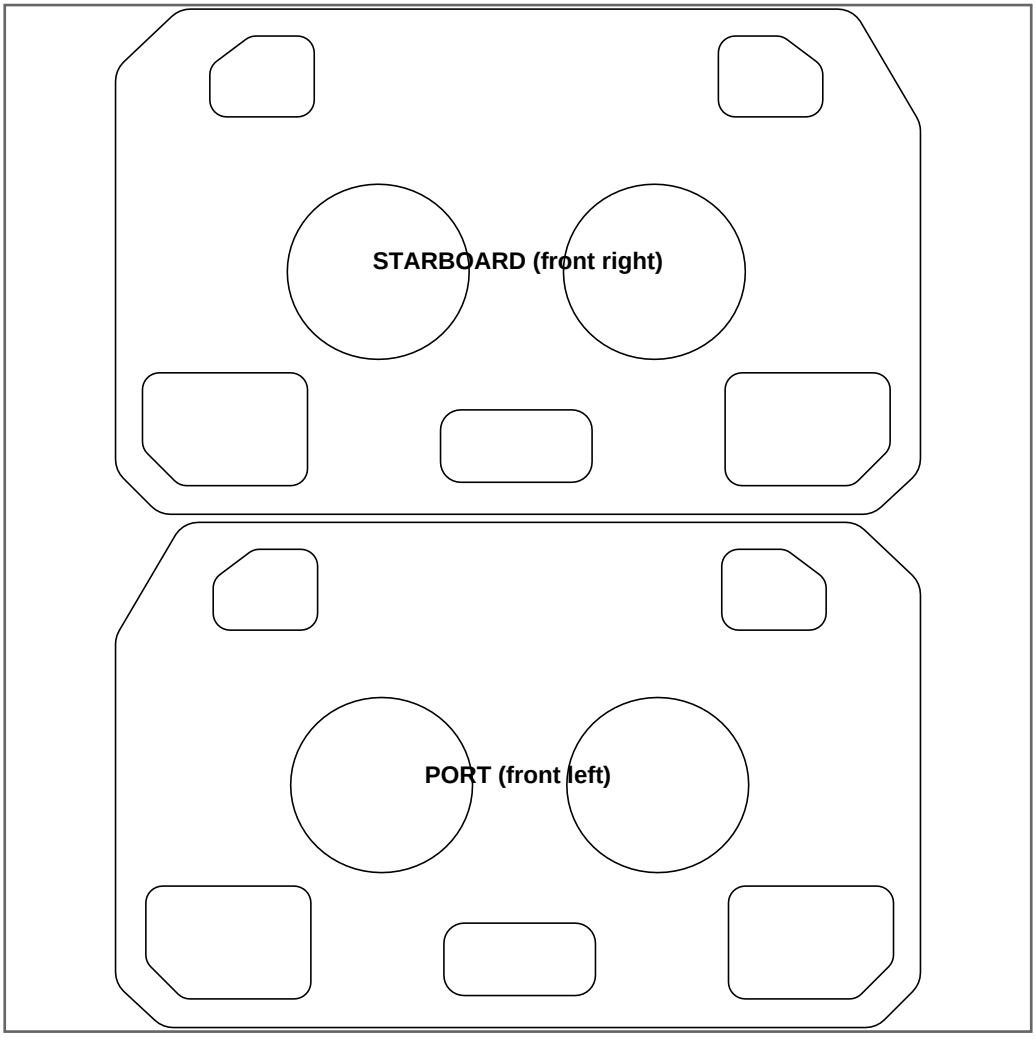
4 Centre-punch every crosshair, then drill: 3.5 mm (M3), 4.5 mm (M4), 5.5 mm (M5) as printed beside each hole. Drill straight (drill press if you have one), slow speed, back the sheet with scrap wood.

5 Drill the 10 mm blue starter hole in each window, then cut the windows with a jigsaw: fine-tooth blade, slow speed, so the HDPE does not melt and re-weld behind the blade. Cut just outside the line.

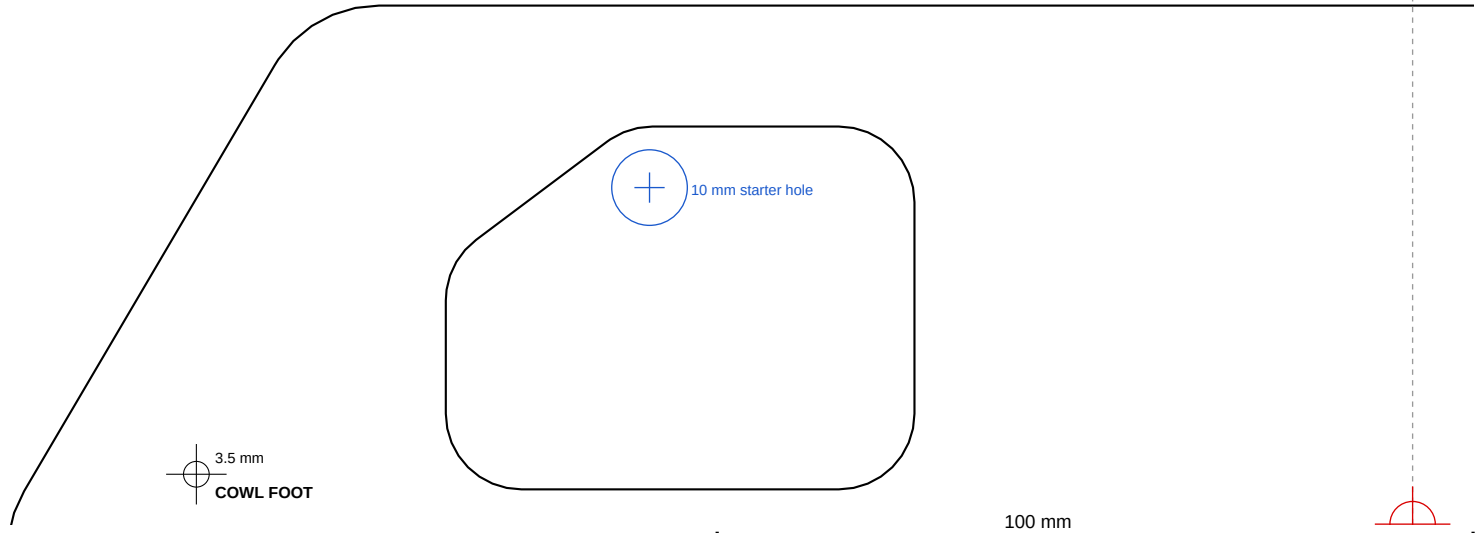
6 Cut the two plate outlines, then file / sand every edge smooth and break the sharp corners.

7 The two plates are mirror images: PORT has the front of the ROV on the left, STARBOARD on the right (seen from outside). Mark each plate's outside face before you peel the template off.

Tools: jigsaw + fine blades, drill + 3.5 / 4.5 / 5.5 / 10 mm bits, centre punch, clamps, file or sandpaper.



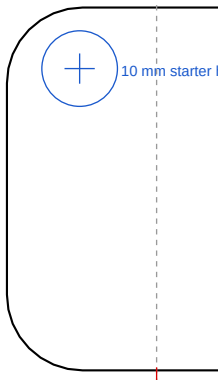
24 x 24 in sheet (609.6 mm), both plates outside face up



901 PORT - tile A1 of 6

seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



3.5 mm
COWL FOOT

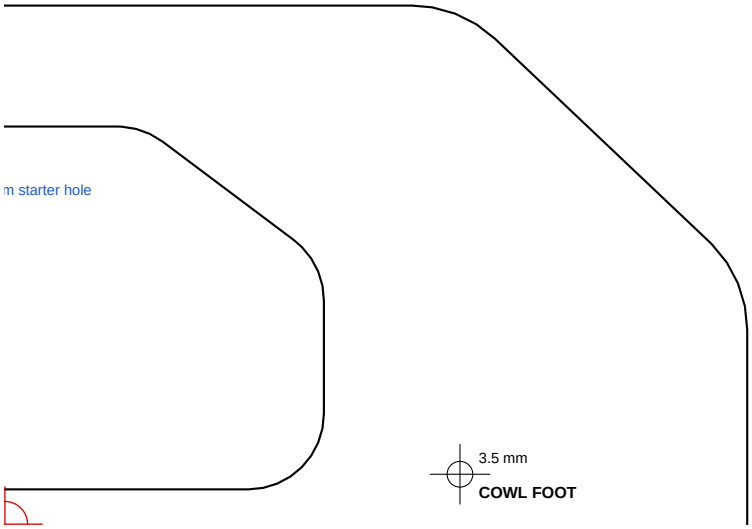
3.5 mm
COWL FOOT

100 mm

901 PORT - tile A2 of 6

seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.

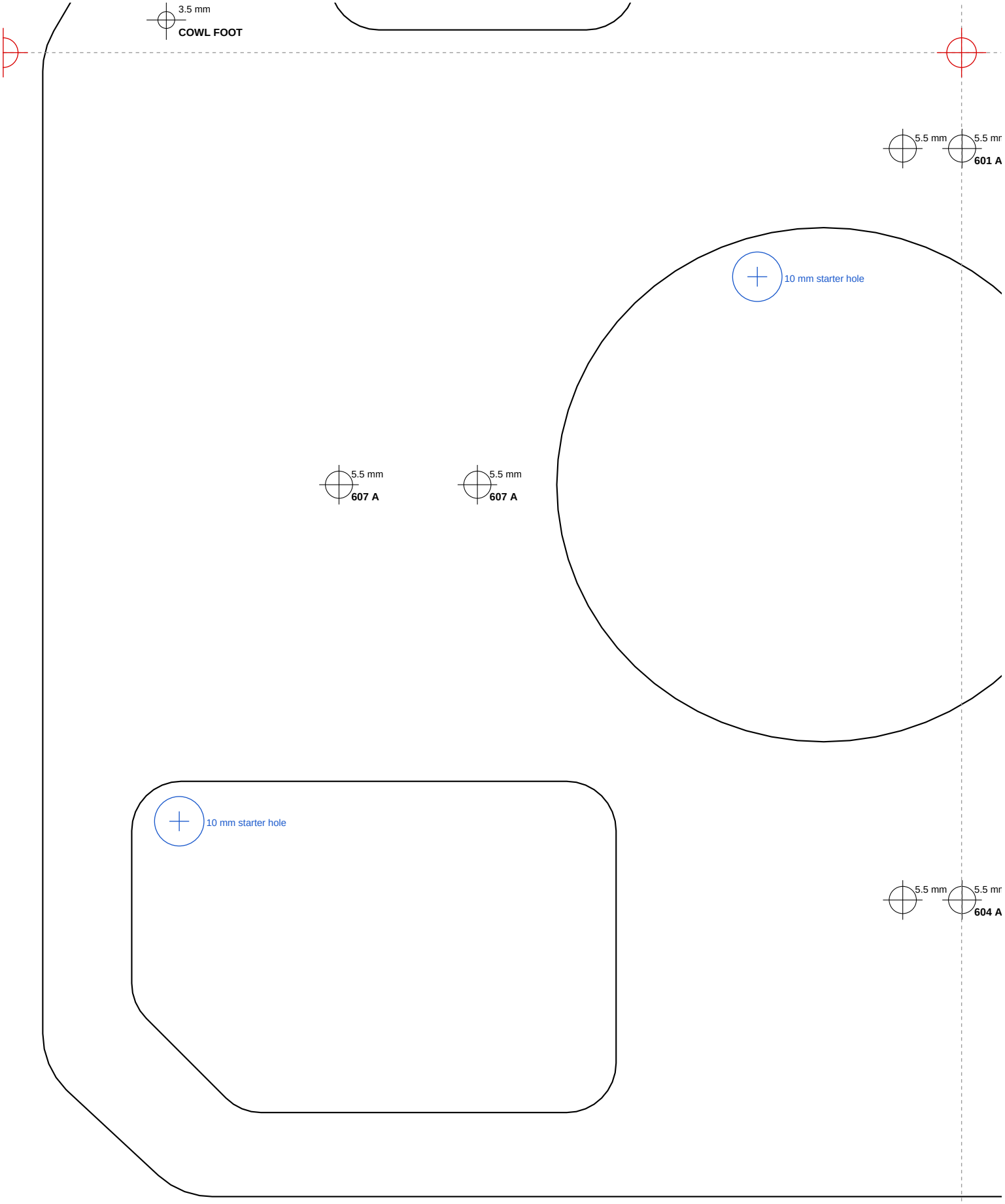


100 mm

901 PORT - tile A3 of 6

seen from OUTSIDE, front of the ROV to the left

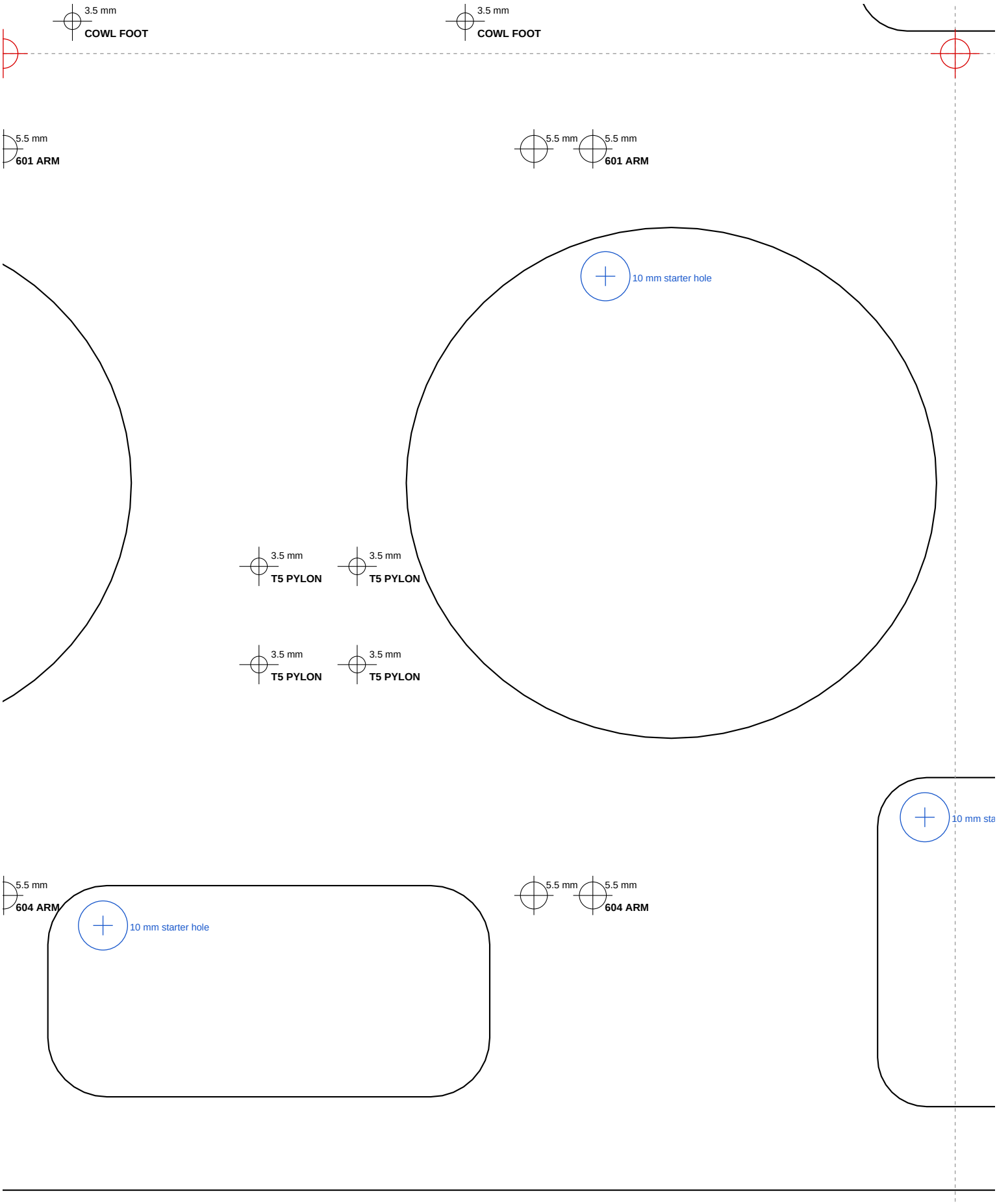
Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



901 PORT - tile B1 of 6

seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.

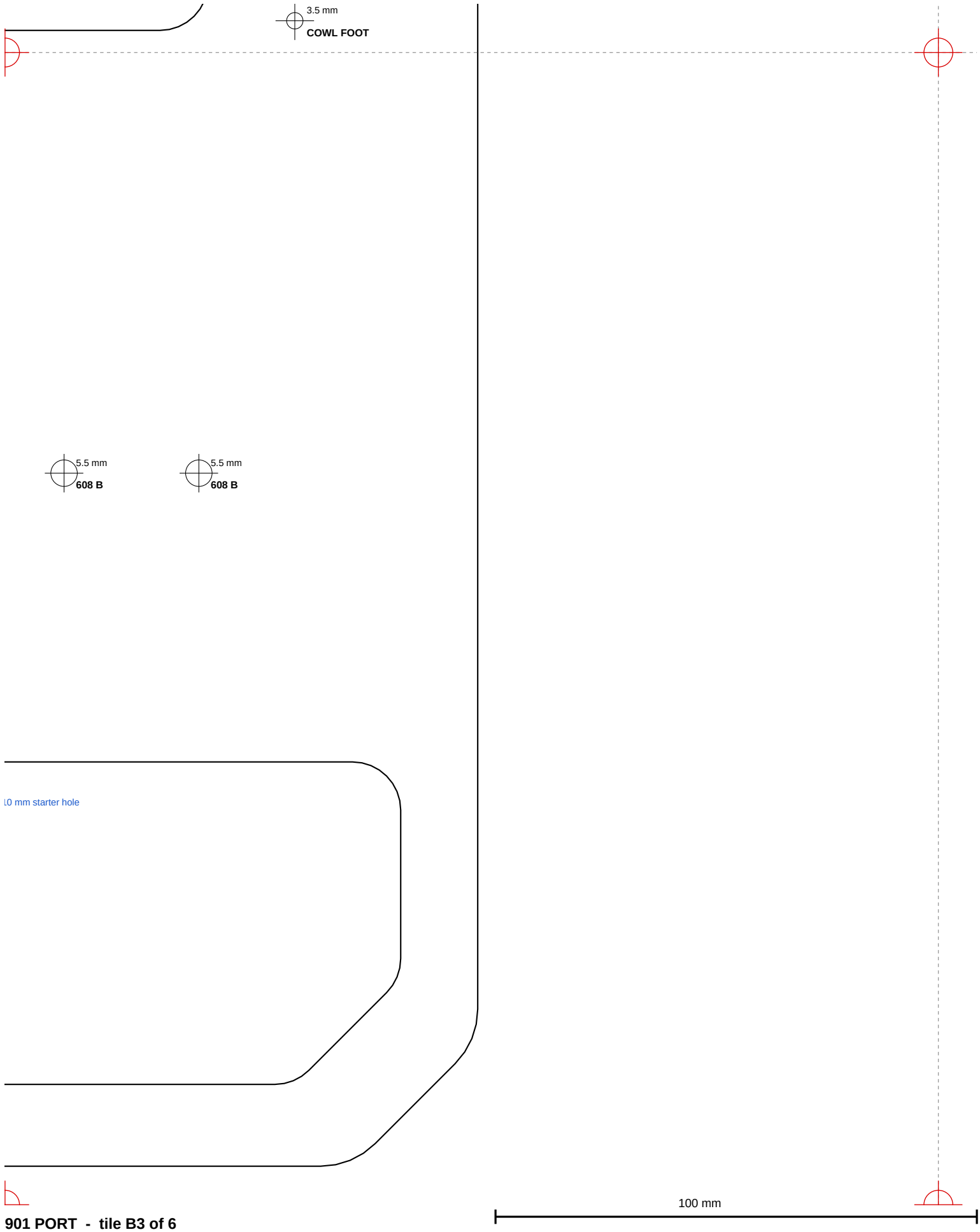


100 mm

901 PORT - tile B2 of 6

seen from OUTSIDE, front of the ROV to the left

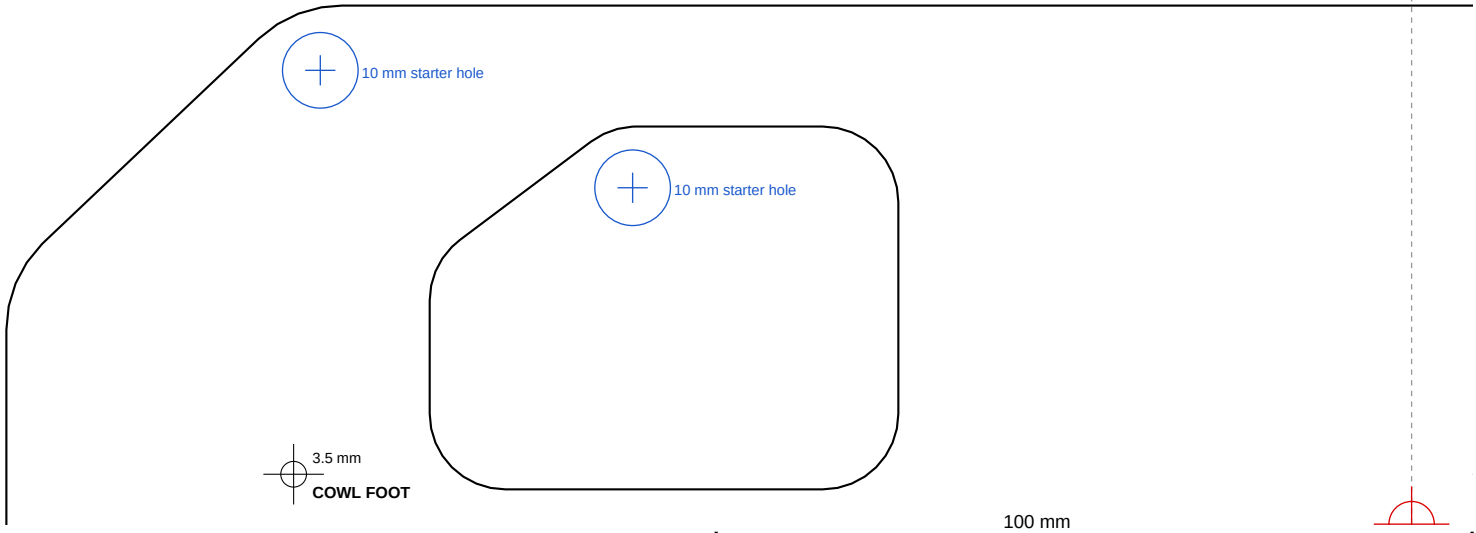
Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



901 PORT - tile B3 of 6

seen from OUTSIDE, front of the ROV to the left

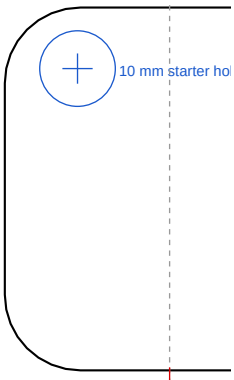
Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



902 STARBOARD - tile A1 of 6

seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



3.5 mm
COWL FOOT

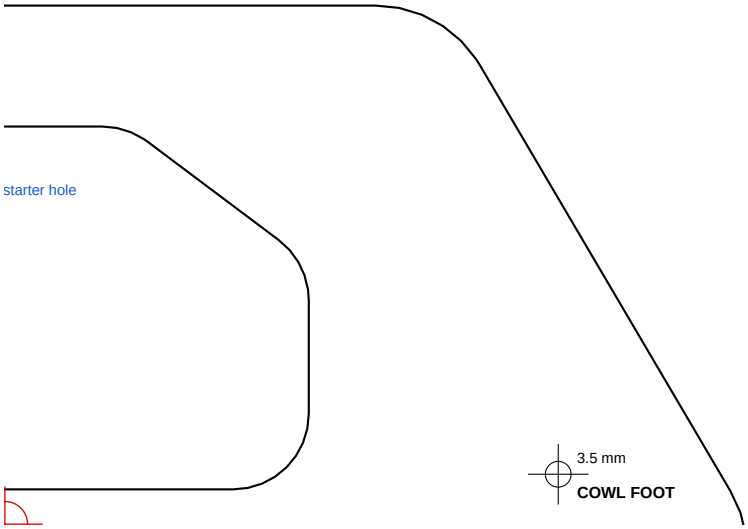
3.5 mm
COWL FOOT

100 mm

902 STARBOARD - tile A2 of 6

seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.

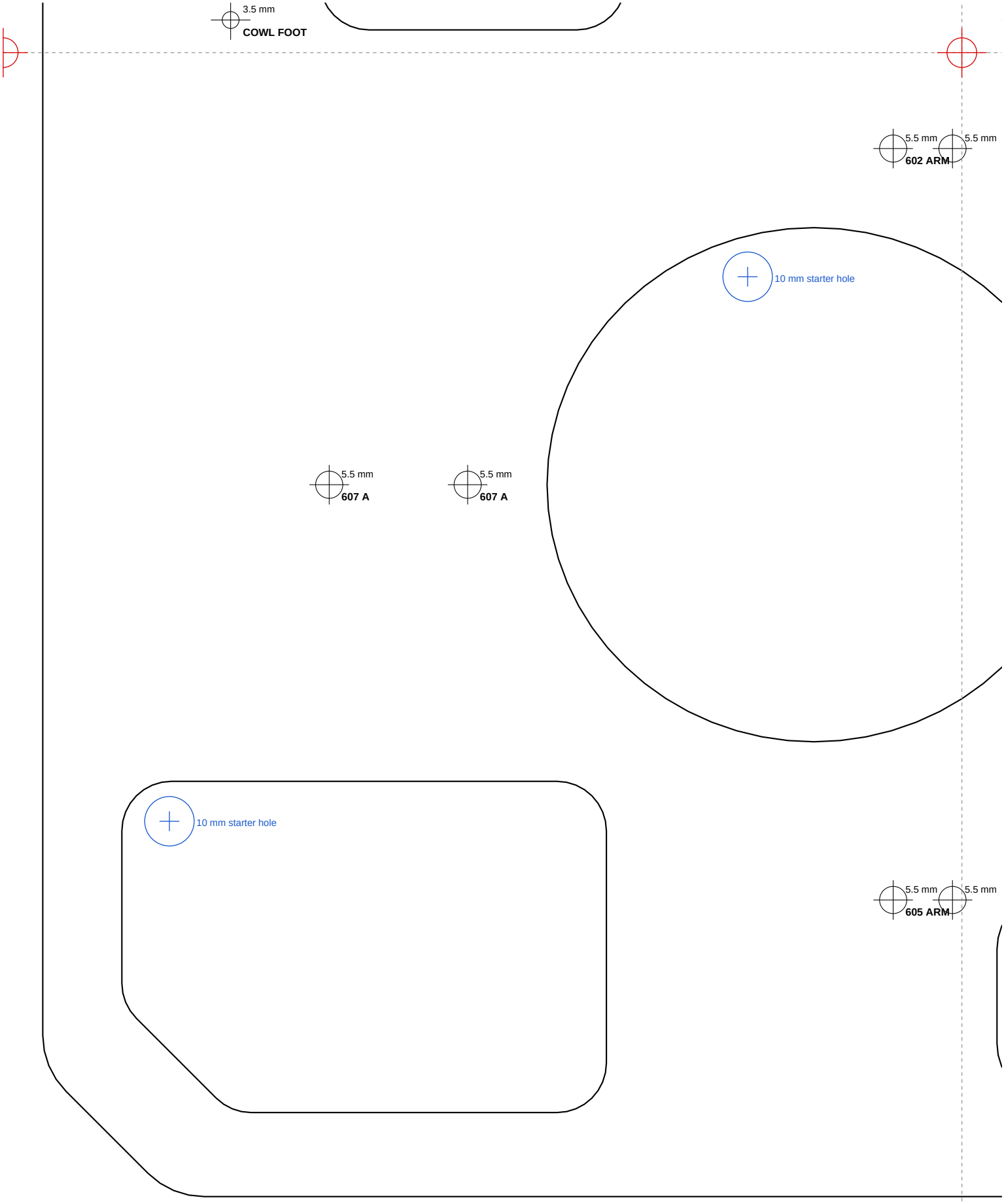


100 mm

902 STARBOARD - tile A3 of 6

seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



902 STARBOARD - tile B1 of 6

seen from OUTSIDE, front of the ROV to the right

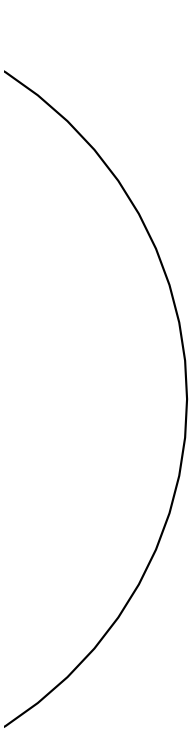
Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.

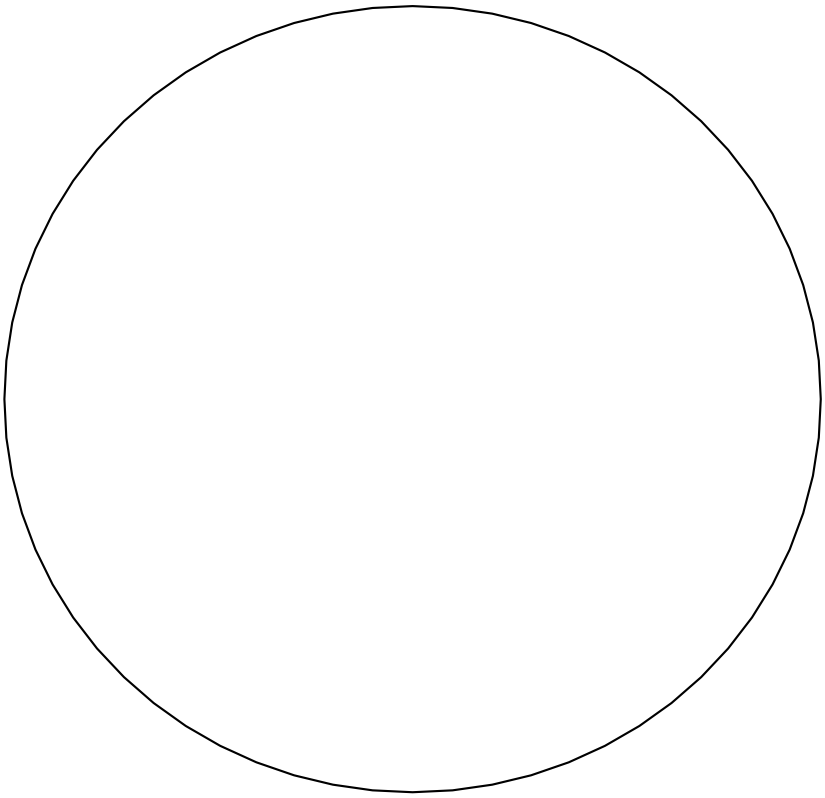
3.5 mm
COWL FOOT

3.5 mm
COWL FOOT

5.5 mm

5.5 mm
5.5 mm
602 ARM





3.5 mm
T6 PYLON

3.5 mm
T6 PYLON

3.5 mm
T6 PYLON

3.5 mm
T6 PYLON

10 mm starte

5.5 mm

10 mm starter hole

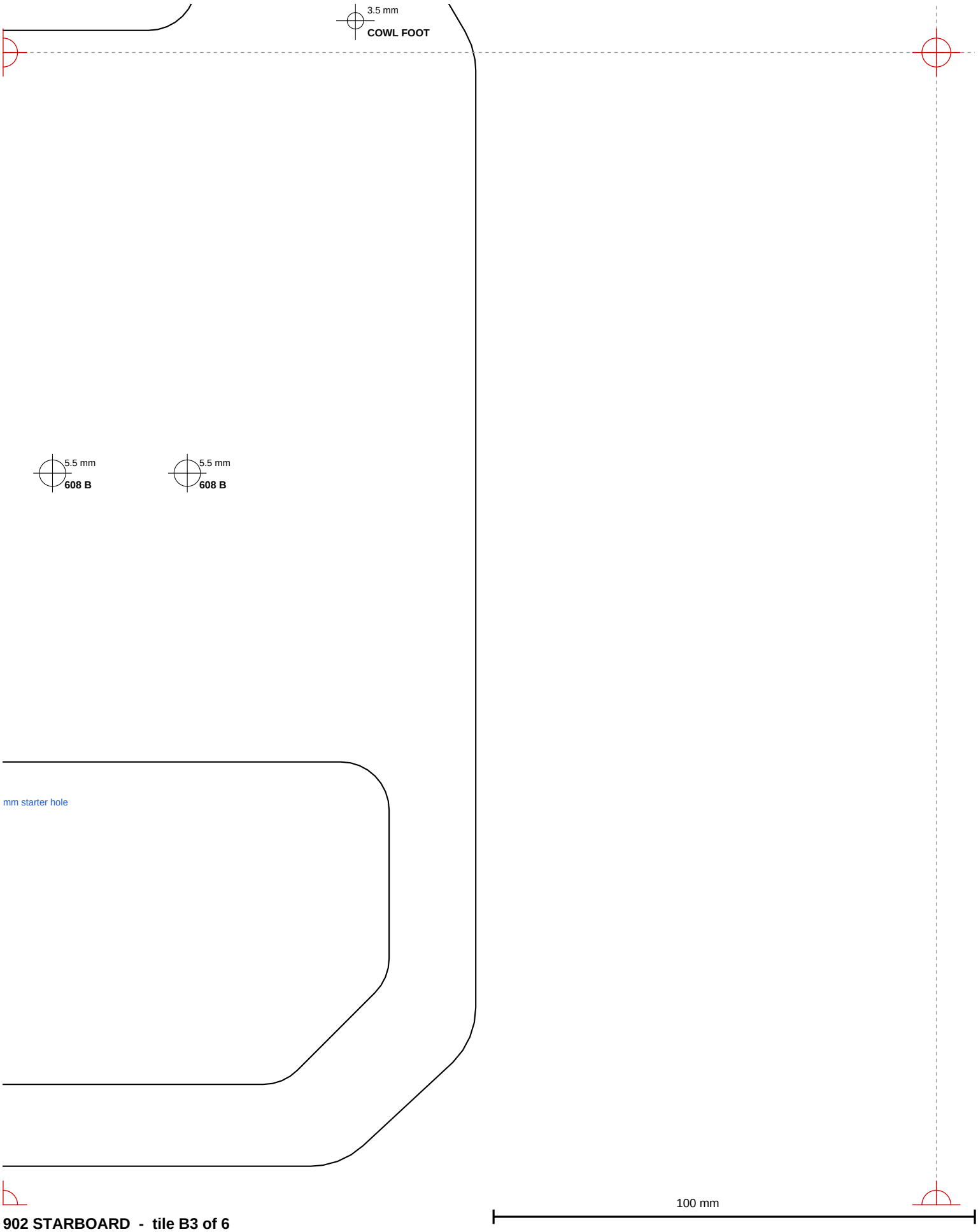
5.5 mm
5.5 mm
605 ARM

100 mm

902 STARBOARD - tile B2 of 6

seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



902 STARBOARD - tile B3 of 6

seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.